

# OptTips

## 50 Modeling Technique Cards

A structured collection of optimization modeling knowledge

**One card, seven components.** Identifier, core idea, applicable patterns, inefficiency symptoms, expert actions, mathematical example, and benchmark guidance. Four cards per page, with a linked index and PDF bookmarks.

### Eight methodological families

|          |                                                               |   |
|----------|---------------------------------------------------------------|---|
| <b>A</b> | Formulation standardization and dimension reduction           | 6 |
| <b>B</b> | Relaxation strengthening and convexification                  | 5 |
| <b>C</b> | Duality, optimality, and equilibrium                          | 4 |
| <b>D</b> | Specialized combinatorial and structured models               | 8 |
| <b>E</b> | Discrete and logical modeling                                 | 4 |
| <b>F</b> | Convex, nonlinear, and structure-preserving modeling          | 9 |
| <b>G</b> | Decomposition and large-scale optimization                    | 6 |
| <b>H</b> | Robust, stochastic, multiobjective, and solver-aware modeling | 8 |

**Scope.** The collection covers formulations, relaxations, and solution procedures. A family is a primary grouping; related techniques can overlap. Each mathematical example illustrates one mechanism within its card, not every variant named in the title. Exact reformulations preserve the stated problem; relaxations provide bounds, and approximations or soft constraints may change it.

**Task coverage.** Test tasks have been constructed for all 50 cards. EfficientOpt evaluates 561 tasks covering 29 cards. A separate extension contains 66 tasks across 22 cards, three per card. T49 appears in both collections, so together they cover all 50. The extension uses native-Python references and is not included in the main Gurobi-based LLM comparisons. Solver support depends on the formulation, not only the card title.

**Examples and evidence.** All cards include symbolic examples. Eight selected execution cases are linked in the index; these illustrate modeling changes without estimating a typical speedup. T10 and T13 compare GPT-5.5 outputs with expert references; the other six compare ordinary and expert references.

**Common comparison rule.** Check feasibility and required outputs in the original problem first. Hold inputs, objectives, hardware, solver settings, and stopping criteria fixed. Measure construction and solving separately, including all iterations or solver calls.

# Technique index

Click a technique title to open its card. A filled marker indicates one of the eight selected execution cases.

| ID  | Technique                                                  | Fam. | p. | ID  | Technique                                                   | Fam. | p. |
|-----|------------------------------------------------------------|------|----|-----|-------------------------------------------------------------|------|----|
| T01 | Equality/inequality equivalent transformation              | A    | 3  | T26 | Complementarity and KKT/MPEC modeling                       | E    | 9  |
| T02 | Auxiliary-variable epigraph/hypograph modeling             | A    | 3  | T27 | Multi-objective and lexicographic optimization              | H    | 9  |
| T03 | Bound tightening and scaling                               | A    | 3  | T28 | Penalty, slack, and feasibility-relaxation modeling         | H    | 9  |
| T04 | LP relaxation of IP/MIP                                    | E    | 3  | T29 | Solver-native modeling constructs                           | H    | 10 |
| T05 | Big- $M$ modeling                                          | E    | 4  | T30 | Warm starts and incumbent injection                         | H    | 10 |
| T06 | Indicator, semicontinuous, and SOS constraints             | E    | 4  | T31 | Algorithm–structure matching                                | H    | 10 |
| T07 | Valid inequalities and cutting planes                      | B    | 4  | T32 | Solver-log-guided diagnosis                                 | H    | 10 |
| T08 | Variable substitution and product linearization            | F    | 4  | T33 | Standard/canonical-form transformation                      | A    | 11 |
| T09 | Piecewise-linear modeling                                  | F    | 5  | T34 | Slack, surplus, and artificial variables                    | A    | 11 |
| T10 | Convex-hull/envelope reformulation and convexification     | B    | 5  | T35 | Variable elimination and dimension reduction                | A    | 11 |
| T11 | Conic reformulation                                        | F    | 5  | T36 | Convexity recognition and DCP modeling                      | F    | 11 |
| T12 | Dual transformation                                        | C    | 5  | T37 | Smooth–nonsmooth composite modeling                         | F    | 12 |
| T13 | Lagrangian relaxation                                      | G    | 6  | T38 | Convex conjugacy and Fenchel duality                        | F    | 12 |
| T14 | Augmented Lagrangian and ADMM                              | G    | 6  | T39 | Subgradient, normal-cone, and optimality-condition modeling | F    | 12 |
| T15 | Block decomposition                                        | G    | 6  | T40 | Projection and proximal-operator modeling                   | F    | 12 |
| T16 | Benders decomposition                                      | G    | 6  | T41 | Variational-inequality and equilibrium-constraint modeling  | C    | 13 |
| T17 | Dantzig–Wolfe decomposition and column generation          | G    | 7  | T42 | Sensitivity and parametric optimization                     | C    | 13 |
| T18 | Network-flow and matching formulations                     | D    | 7  | T43 | Dynamic programming and Bellman recursion                   | G    | 13 |
| T19 | Covering, partitioning, packing, and knapsack formulations | D    | 7  | T44 | Transportation and network-simplex structure                | D    | 13 |
| T20 | Scheduling and temporal formulations                       | D    | 7  | T45 | Constraint qualifications and KKT validity                  | C    | 14 |
| T21 | Routing and path formulations                              | D    | 8  | T46 | Interior-point-friendly modeling                            | F    | 14 |
| T22 | Robust counterpart reformulation                           | H    | 8  | T47 | Matrix-variable and low-rank modeling                       | D    | 14 |
| T23 | Stochastic programming and scenario decomposition          | H    | 8  | T48 | Cover, clique, subtour, and capacity cuts                   | B    | 14 |
| T24 | Preprocessing and dominance elimination                    | B    | 8  | T49 | Linking constraints and activation strengthening            | B    | 15 |
| T25 | Symmetry breaking                                          | D    | 9  | T50 | Tensor-variable and low-rank modeling                       | D    | 15 |

Selected benchmark cases

16

Each identifier is stable across the source library, the manuscript, and this collection.

## Technique cards

T01–T04

## FAMILY A

## T01 Equality/inequality equivalent transformation

**Core idea:** Rewrite constraints into equivalent algebraic forms while keeping the original domains and objective. Expose linear structure before adding auxiliary variables.

**Applicable patterns:** Balances, intervals, and fixed-ratio restrictions appear in LP, QP, and MIP models. Clearing a denominator requires a proven nonzero sign.

**Inefficiency symptoms:** Duplicate rows, inconsistent inequality directions, or symbolic divisions obscure constraints that admit simpler algebra.

**Expert actions:** Normalize signs and clear denominators only on a justified domain. Preserve equalities, bounds, and variable types; check both implication directions.

**Mathematical example:** Let  $a, d, \alpha, \beta$  be fixed coefficients. The original domain guarantees  $d^\top x + \beta \geq \delta$  for some  $\delta > 0$ .

## Fractional inequality

$$\frac{a^\top x}{d^\top x + \beta} \leq \alpha$$

$$d^\top x + \beta \geq \delta$$

## Equivalent linear row

$$(a - \alpha d)^\top x \leq \alpha \beta$$

$$d^\top x + \beta \geq \delta$$

A negative denominator reverses the inequality; an unknown sign needs separate treatment. See T33–T34 for canonical forms and slacks.

**Benchmark guidance:** Verify feasible-set equivalence under the stated domain and retain the objective. Compare model class, rows, sparsity, and construction cost.

**Constructed tasks:** 32 in EfficientOpt.

Source: OptTips / T01.

## FAMILY A

## T03 Bound tightening and scaling

**Core idea:** Derive valid tighter bounds and rescale quantities consistently. These strengthen conditional constraints and envelopes while reducing numerical scale differences.

**Applicable patterns:** Production, inventory, scheduling, and energy models often expose capacity or temporal bounds. Every tightened bound must follow from the original task.

**Inefficiency symptoms:** Available limits are omitted, deactivation constants remain loose, or incompatible units create extreme coefficient ranges.

**Expert actions:** Propagate capacities, balances, and time bounds; transform variables, objective coefficients, and units together. Record the inverse map and preserve integer lattices.

**Mathematical example:** For continuous  $x$ , capacity implies  $x \leq 2$ . Set  $x = 2z$  to normalize the feasible interval.

## Loose bound and capacity

$$\begin{array}{ll} \min & cx \\ \text{s.t.} & 1000x \leq 2000, \\ & 0 \leq x \leq 10^6 \end{array}$$

## Tightened and scaled

$$\begin{array}{ll} \min & 2cz \\ \text{s.t.} & 0 \leq z \leq 1 \end{array}$$

Recover  $x = 2z$  and check original-unit tolerances. If  $x$  is integer,  $z$  lies on a half-integer lattice, not simply the integers.

**Benchmark guidance:** Check bound implications and recovered feasibility in original units. Compare relaxation bounds, coefficient ranges, presolve, and solver work at consistent accuracy.

**Selected case:** T03\_001 (p. 16)

OptTips / T03 · 19 EfficientOpt tasks.

## FAMILY A

## T02 Auxiliary-variable epigraph/hypograph modeling

**Core idea:** Expose maximum, minimum, or absolute-value structure with an auxiliary variable. Objective monotonicity must enforce its intended value.

**Applicable patterns:** Worst-case costs, peak loads, fairness, and absolute-deviation regression. Convex epigraphs and concave hypographs preserve convexity under compatible operations.

**Inefficiency symptoms:** Nested objectives hide structure; host-language maximum or absolute-value calls may not accept symbolic decisions. Incorrect auxiliary constraints can change the objective.

**Expert actions:** Add bounds with the correct direction. For absolute deviations, use epigraphs or positive/negative parts under an appropriate minimizing penalty.

**Mathematical example:** Let  $x \in X$  and each  $g_i(x)$  be finite on the feasible domain. The objective minimizes the largest component.

## Maximum objective

$$\min_{x \in X} \max_{i=1, \dots, m} g_i(x)$$

## Epigraph formulation

$$\begin{array}{ll} \min_{x, t} & t \\ \text{s.t.} & g_i(x) \leq t \quad (\forall i), \\ & x \in X \end{array}$$

At a finite optimum, minimization makes  $t = \max_i g_i(x)$ . Convex  $g_i$  give a convex epigraph; equivalence itself does not require convexity.

**Benchmark guidance:** Compare valid direct and auxiliary formulations. Check projected feasibility and objective equality, then inspect model class, auxiliary dimensions, and solver work.

**Constructed tasks:** 15 in EfficientOpt.

Source: OptTips / T02.

## FAMILY E

## T04 LP relaxation of IP/MIP

**Core idea:** Relax integrality to assess formulation strength and expose troublesome fractional patterns. The relaxation supplies a bound, generally not an integer task solution.

**Applicable patterns:** Apply to binary linear knapsack, facility-location, routing, scheduling, and packing models. The relaxation must contain every original integer solution with unchanged objective.

**Inefficiency symptoms:** Branching grows rapidly with instance size, or the relaxation objective remains distant from feasible incumbents. Fractional activation and assignment patterns reveal weak links.

**Expert actions:** Inspect fractional patterns and add justified strengthening. Recover integer feasibility through a valid heuristic, repair, or integer solve rather than unconditional rounding.

**Mathematical example:** For feasible binary minimization,  $z_{LP} \leq z_{IP}$ . The relaxation provides a lower bound, not necessarily an integer solution.

## Integer model

$$\begin{array}{ll} z_{IP} = \min & c^\top x \\ \text{s.t.} & Ax \geq b, \\ & x \in \{0, 1\}^n \end{array}$$

## Linear relaxation

$$\begin{array}{ll} z_{LP} = \min & c^\top x \\ \text{s.t.} & Ax \geq b, \\ & 0 \leq x \leq 1 \end{array}$$

Maximization reverses the bound direction. Integer recovery needs verification; the source gap is  $|z_{IP} - z_{LP}| / (1 + |z_{IP}|)$ .

**Benchmark guidance:** Report relaxation direction and fractional patterns. Assess strengthened integer formulations separately, using verified feasible incumbents and an explicit normalized relaxation-gap definition.

**Constructed tasks:** 25 in EfficientOpt.

Source: OptTips / T04.

## Technique cards

T05–T08

## FAMILY E

T05 Big- $M$  modeling

**Core idea:** Encode conditional rows with finite deactivation constants derived from valid bounds. Oversized constants weaken relaxations; underestimated constants may remove feasible decisions.

**Applicable patterns:** Apply to conditional capacities, facility switches, nonoverlap, commitment, and batch activation. Each implication requires bounded inactive-branch activity and a correctly directed condition.

**Inefficiency symptoms:** Unrelated rows share arbitrary large constants, broadening coefficient ranges and weakening fractional activations. Insufficient constants silently invalidate the intended logical condition.

**Expert actions:** Tighten bounds, maximize each conditional row activity, and derive a valid nonnegative constant. Consider equivalent native indicators or semicontinuous constructs where supported.

**Mathematical example:** Let  $y$  be binary and  $l \leq x \leq u$  finite valid bounds. Encode  $y = 1 \Rightarrow a^\top x \leq b$ .

## Valid conservative constant

$$a^\top x \leq b + \overline{M}(1 - y), \\ l \leq x \leq u, \quad y \in \{0, 1\}$$

## Bound-derived constant

$$U_a = \max_{l \leq x \leq u} a^\top x, \\ M^* = \max\{0, U_a - b\}, \\ a^\top x \leq b + M^*(1 - y)$$

Assume  $\overline{M} \geq M^*$ . The box-derived constant is valid; other common constraints may permit a smaller constant.

**Benchmark guidance:** Compare valid conservative and derived constants. Check inactive-branch coverage and binary feasibility, then inspect coefficient ranges, relaxation bounds, and solver work.

**Constructed tasks:** 19 in EfficientOpt.

Source: OptTips / T05.

## FAMILY E

## T06 Indicator, semicontinuous, and SOS constraints

**Core idea:** Express logical and discrete structure using supported native constructs. This exposes intended semantics and permits specialized solver handling without changing the decisions.

**Applicable patterns:** Use indicators for implications, semicontinuous domains for zero-or-batch quantities, SOS1 for exclusion, and SOS2 for adjacent interpolation. Check solver-specific domain support.

**Inefficiency symptoms:** Manual conditional expansions add oversized constants or excessive rows; pairwise exclusions grow rapidly. Interpolation may omit adjacency, or simple domains require unnecessary activations.

**Expert actions:** Choose supported indicators, semicontinuous domains, SOS1/SOS2, or enforcement literals. Preserve separate activation costs and coupling roles, and document solver compatibility and semantics.

**Mathematical example:** Assume  $0 < L \leq U$  and activation used only in these links. Production is either zero or within its positive batch interval.

## Explicit activation

$$Ly \leq x \leq Uy, \\ y \in \{0, 1\}$$

## Native semicontinuous domain

$$x \in \{0\} \cup [L, U]$$

Recover  $y = 0$  at zero and  $y = 1$  otherwise. Separate activation costs or coupling roles still need representation; semicontinuity remains discrete.

**Benchmark guidance:** Compare valid explicit and native encodings. Verify decision correspondence, domains, and objectives; examine rows, internal reformulations, and solver work without assuming uniform improvement.

**Selected case:** T06\_010 (p. 17)

OptTips / T06 · 21 EfficientOpt tasks.

## FAMILY B

## T07 Valid inequalities and cutting planes

**Core idea:** Add inequalities valid for all original integer solutions to remove fractional points. Install them initially or generate cutting planes through separation.

**Applicable patterns:** Apply to knapsack, routing, facility location, packing, and network design. Justify each inequality against the complete integer feasible set and variable domains.

**Inefficiency symptoms:** Fractional relaxation solutions persist, or an incumbent is found long before optimality is proved. Weak capacity and connectivity descriptions leave excessive freedom.

**Expert actions:** Derive cover, clique, connectivity, capacity, or flow-cover inequalities. Distinguish relaxation-strengthening user cuts from lazy constraints needed to define original task feasibility.

**Mathematical example:** For nonnegative knapsack weights, a cover  $C$  satisfies  $\sum_{i \in C} a_i > b$ : selecting every cover item is infeasible.

## Knapsack formulation

$$\sum_{i=1}^n a_i x_i \leq b, \\ x \in \{0, 1\}^n$$

## Additional cover inequality

$$\sum_{i \in C} x_i \leq |C| - 1$$

Retain the knapsack row: cover cuts preserve binary solutions but may exclude fractional points. See T48 for specialized cuts and T49 for activation links.

**Benchmark guidance:** Compare the same integer model with and without valid inequalities. Check objective preservation, then inspect relaxation bounds, nodes, cut counts, and separation overhead.

**Constructed tasks:** 16 in EfficientOpt.

Source: OptTips / T07.

## FAMILY F

## T08 Variable substitution and product linearization

**Core idea:** Introduce an auxiliary product variable and a domain-valid linear description. Binary factors permit exact formulations; continuous products generally require relaxation or nonlinear treatment.

**Applicable patterns:** Apply to switched quantities, conjunctions, fixed charges, and activation terms. Exact binary–continuous linearization requires finite factor bounds and inequalities appropriate to their signs.

**Inefficiency symptoms:** A binary product is sent to a general nonconvex solver despite exact linearization. Missing bounds weaken or invalidate the proposed auxiliary formulation.

**Expert actions:** Add every required bounding inequality. Use AND constraints for binary factors; distinguish exact binary linearization from McCormick relaxation when both factors remain continuous.

**Mathematical example:** Let  $y \in \{0, 1\}$  and  $0 \leq x \leq U$  with finite  $U \geq 0$ . The auxiliary  $z$  must equal the switched quantity  $xy$ .

## Binary–continuous product

$$z = xy, \\ 0 \leq x \leq U, \quad y \in \{0, 1\}$$

## Exact linearization

$$z \leq Uy, \\ z \leq x, \\ z \geq x - U(1 - y), \\ z \geq 0$$

Retain factor bounds and integrality. Binary–binary AND also requires  $0 \leq z \leq 1$  alongside its three standard inequalities.

**Benchmark guidance:** Compare valid direct and linearized models. Check products at binary decisions, preserve original constraints, and inspect model class, relaxation strength, and solver work.

**Constructed tasks:** 46 in EfficientOpt.

Source: OptTips / T08.

## Technique cards

T09–T12

## FAMILY F

## T09 Piecewise-linear modeling

**Core idea:** Represent piecewise-linear structure using curvature and objective monotonicity. Epigraphs can avoid binaries; graph equality generally requires valid segment selection.

**Applicable patterns:** Tariffs, costs, efficiency curves, and taxes use ordered breakpoints. Discontinuous steps require explicit boundary conventions rather than continuous interpolation.

**Inefficiency symptoms:** Host-language branches obscure symbolic decisions, interpolation mixes nonadjacent segments, or unnecessary binaries encode a convex objective.

**Expert actions:** Use epigraph rows only when lowering the value variable preserves feasibility and cannot worsen the objective. Otherwise retain valid graph structure.

**Mathematical example:** Let  $x_0 < \dots < x_m$  and let  $f$  linearly interpolate  $(x_k, v_k)$ . SOS2 gives this continuous graph exactly.

## Specified piecewise-linear graph

$$v = f(x), \quad x \in [x_0, x_m], \\ f(x_k) = v_k \quad (k = 0, \dots, m)$$

## Adjacent convex combination

$$x = \sum_{k=0}^m \lambda_k x_k, \\ v = \sum_{k=0}^m \lambda_k v_k, \\ \sum_k \lambda_k = 1, \quad \lambda \geq 0, \\ \text{SOS2}(\lambda)$$

SOS2 permits at most two adjacent nonzero weights. Convexity alone does not justify replacing  $v = f(x)$  by  $v \geq f(x)$  in arbitrary surrounding constraints.

**Benchmark guidance:** Check endpoints, segment values, and auxiliary uses. Compare valid encodings of the same curve through binaries, dimensions, and solver work.

**Selected case:** T09\_008 (p. 17)

OptTips / T09 · 8 EfficientOpt tasks.

## FAMILY B

## T10 Convex-hull/envelope reformulation and convexification

**Core idea:** Use convex hulls, envelopes, or extended formulations to strengthen continuous relaxations. Extra variables can describe feasible combinations more faithfully.

**Applicable patterns:** Bounded products, disjunctions, fixed-charge convex costs, commitment, and sparse selection admit useful convex descriptions.

**Inefficiency symptoms:** Fractional solutions exploit loose deactivation constants or products; weak bounds slow certification of optimality.

**Expert actions:** Derive domain-valid envelopes, ideal disjunctions, or closed perspectives. Distinguish exact reformulation from relaxation and retain necessary original constraints.

**Mathematical example:** For finite factor bounds, these inequalities describe the convex hull of a single bilinear graph over its box.

## Bilinear graph

$$z = xy, \\ l_x \leq x \leq u_x, \\ l_y \leq y \leq u_y$$

## McCormick envelope

$$z \geq l_x y + l_y x - l_x l_y, \\ z \geq u_x y + u_y x - u_x u_y, \\ z \leq u_x y + l_y x - u_x l_y, \\ z \leq l_x y + u_y x - l_x u_y$$

Continuous envelope points need not satisfy  $z = xy$ . Termwise envelopes do not generally give the convex hull of a coupled model.

**Benchmark guidance:** Compare dimensions and relaxation bounds. Separately verify original feasibility and objective quality; a stronger bound is not a feasible task solution.

**Selected case:** T10\_011 (p. 16)

OptTips / T10 · 30 EfficientOpt tasks.

## FAMILY F

## T11 Conic reformulation

**Core idea:** Expose recognized convex nonlinear structure through exact cone representations. This makes curvature and domains explicit while enabling specialized algorithms without arbitrary discretization.

**Applicable patterns:** Use second-order, rotated, semidefinite, exponential, or power cones in robust, portfolio, power, and entropy models. Match domains and available solver support.

**Inefficiency symptoms:** Norms, exponential structure, or positive-semidefinite constraints are hidden in generic nonlinear code. Coarse piecewise approximations enlarge the model while changing its relationships.

**Expert actions:** Identify the cone and required auxiliaries, retaining domain restrictions and scaling factors. Verify algebraic equivalence, perspective closure conventions, and matrix definiteness requirements.

**Mathematical example:** For real  $x$  and  $y$ ,  $z \geq 0$ , the rotated quadratic inequality has the displayed exact second-order-cone representation.

## Rotated quadratic inequality

$$x^2 \leq yz, \\ y \geq 0, \quad z \geq 0$$

## Second-order-cone form

$$\left\| \begin{pmatrix} 2x \\ y - z \end{pmatrix} \right\|_2 \leq y + z, \\ y \geq 0, \quad z \geq 0$$

The factor two is essential. Squaring is valid since  $y + z \geq 0$ ; exponential perspectives at zero denominators instead require closed-cone definitions.

**Benchmark guidance:** Compare valid direct and exact conic models. Verify objectives and original-variable feasibility, then inspect solver status, iterations, residuals, and formulation dimensions.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T11.

## FAMILY C

## T12 Dual transformation

**Core idea:** Replace a suitable subproblem by its dual to expose an alternative formulation. Strong duality, correct signs, outer quantifiers, and primal recovery are essential.

**Applicable patterns:** Apply to LPs, budgeted robust costs, shadow prices, and suitable convex inner problems. Check feasibility, boundedness, and strong-duality conditions before substitution.

**Inefficiency symptoms:** Uncertainty is explicitly enumerated or inner optimization remains hidden. Attempted dual substitutions omit feasibility requirements, reverse signs, or assume unjustified strong duality.

**Expert actions:** Derive consistent primal and dual models, verify assumptions, and preserve outer logic. Recover required primal decisions or supply appropriate accompanying optimality certificates.

**Mathematical example:** The minimization LP has nonnegative variables and lower-bounding rows. If primal and dual are feasible, finite optimal values coincide.

## Primal linear program

$$\min \quad c^\top x \\ \text{s.t.} \quad Ax \geq b, \\ x \geq 0$$

## Dual linear program

$$\max \quad b^\top y \\ \text{s.t.} \quad A^\top y \leq c, \\ y \geq 0$$

A dual optimum is not a primal decision vector. Recover required outputs and verify feasibility; nested substitutions must preserve outer quantifiers.

**Benchmark guidance:** Compare valid primal and justified dual counterparts. Check objective agreement and recovered primal feasibility, then assess dimensions and total effort including recovery.

**Selected case:** T12\_004 (p. 16)

OptTips / T12 · 6 EfficientOpt tasks.

## Technique cards

T13–T16

## FAMILY G

## T13 Lagrangian relaxation

**Core idea:** Dualize shared constraints to separate local subproblems and optimize a bound. Relaxed decisions require separate recovery of original coupling feasibility.

**Applicable patterns:** Few resource rows couple manageable blocks in commitment, timetabling, networks, or supply chains; local domains may include integers.

**Inefficiency symptoms:** Shared capacities make a monolithic formulation difficult although each uncoupled block is simple.

**Expert actions:** Choose coupling rows, derive multiplier signs, and solve local subproblems. Update multipliers through justified methods and maintain feasible primal recovery.

**Mathematical example:** Local  $x_k \in X_k$  interact only through the displayed resource constraint; use multipliers  $\lambda \geq 0$ .

## Coupled blocks

$$p^* = \inf_{\text{s.t.}} \sum_k f_k(x_k) \\ \sum_k A_k x_k \leq b, \\ x_k \in X_k \quad (\forall k)$$

## Separable Lagrangian bound

$$q_k(\lambda) = \inf_{x_k \in X_k} \{f_k(x_k) + \lambda^\top A_k x_k\}, \\ q(\lambda) = \sum_k q_k(\lambda) - \lambda^\top b, \\ \sup_{\lambda \geq 0} q(\lambda) \leq p^*$$

Integer or nonconvex blocks may leave a positive duality gap. Equality requires additional assumptions; primal recovery remains separate.

**Benchmark guidance:** Report lower bounds, recovered feasible objectives, coupling violations, and all subproblem/coordination effort. Do not score infeasible relaxed decisions as task solutions.

**Selected case:** T13\_006 (p. 17)

OptTips / T13 · 15 EfficientOpt tasks.

## FAMILY G

## T14 Augmented Lagrangian and ADMM

**Core idea:** Coordinate two separable blocks through augmented penalties and multiplier updates. Guarantees depend on objective, solvability, and update assumptions.

**Applicable patterns:** Continuous convex consensus, energy, control, and learning problems with linear coupling.

**Inefficiency symptoms:** Centralized storage or coordination is difficult; independently optimized blocks violate shared consistency.

**Expert actions:** Check proper closed convex objectives, a saddle point, and solvable subproblems. Select penalty and scaled-dual updates consistently.

Mathematical example: For  $\rho > 0$ , use  $L_\rho = f(x) + g(z) + (\rho/2)\|Ax + Bz - c + u\|_2^2 - (\rho/2)\|u\|_2^2$ ;  $u$  is scaled dual. Primal-update constants are omitted.

## Two-block coupled problem

$$\min_{x,z} f(x) + g(z) \\ \text{s.t.} \quad Ax + Bz = c$$

## Scaled ADMM updates

$$x^{k+1} \in \arg \min_x L_\rho(x, z^k, u^k), \\ z^{k+1} \in \arg \min_z L_\rho(x^{k+1}, z, u^k), \\ u^{k+1} = u^k + Ax^{k+1} + Bz^{k+1} - c$$

Use the updated first block in the second update. Standard guarantees do not automatically extend to arbitrary multiblock, nonconvex, or inexact variants.

**Benchmark guidance:** Specify absolute/relative primal and dual residual tolerances. Compare objective and feasibility accuracy, counting every local solve, iteration, and communication.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T14.

## FAMILY G

## T15 Block decomposition

**Core idea:** Expose local structure while retaining all coordinating decisions. Incomplete coordination or truncated horizons may introduce approximation.

**Applicable patterns:** Time, scenario, region, or variable blocks in scheduling, stochastic programming, and network design.

**Inefficiency symptoms:** A monolithic model hides local structure, or independent blocks omit shared resources and boundary dependencies.

**Expert actions:** Separate local and coordinating variables. Preserve coupling and boundary states; disclose horizon truncation or aggregation approximations.

**Mathematical example:** Fixing  $y \in Y$  gives local subproblems. Assume each feasible subproblem attains a finite optimum.

## Monolithic block model

$$\min \phi(y) + \sum_k f_k(x_k) \\ \text{s.t.} \quad A_k x_k + B_k y \leq b_k \quad (\forall k), \\ x_k \in X_k, \quad y \in Y$$

## Value-function coordination

$$\min_{y \in Y} \phi(y) + \sum_k Q_k(y), \\ Q_k(y) = \inf \{f_k(x_k) : x_k \in X_k, \\ A_k x_k + B_k y \leq b_k\}$$

Infeasible blocks have value  $+\infty$ . Exact value functions preserve this model; optimizing the shared  $y$  remains necessary.

**Benchmark guidance:** Verify assembled decisions against every original constraint. Compare accuracy and total master, subproblem, and communication effort.

**Constructed tasks:** 24 in EfficientOpt.

Source: OptTips / T15.

## FAMILY G

## T16 Benders decomposition

**Core idea:** Separate design decisions from continuous recourse. A master coordinates designs through fixed-design dual optimality and feasibility cuts.

**Applicable patterns:** Facility location, network design, capacity planning, and stochastic models with linear recourse and explicit feasibility/boundedness handling.

**Inefficiency symptoms:** Many operating variables share a small design vector; fixing designs exposes manageable LPs.

**Expert actions:** Initialize the recourse surrogate with a valid lower bound or dual-feasible cut. Separate violated cuts and recover operating decisions.

**Mathematical example:** For nonempty  $P = \{\pi \geq 0 : A^\top \pi \leq c\}$ , impose the cuts for every extreme point  $\pi \in V$  and extreme ray  $r \in R$ .

## Design and linear recourse

$$\min f^\top y + c^\top x \\ \text{s.t.} \quad Ax \geq b - By, \\ x \geq 0, \quad y \in Y$$

## Benders master

$$\min f^\top y + \eta \\ \text{s.t.} \quad \eta \geq \pi^\top (b - By), \\ r^\top (b - By) \leq 0, \\ y \in Y$$

Rays satisfy  $r \geq 0$ ,  $A^\top r \leq 0$ . Complete cuts represent recourse; restricted masters require separation. Do not impose  $\eta \geq 0$  without justification.

**Benchmark guidance:** Validate cuts and recovered feasibility; compare bound gaps, iterations, and total master/subproblem effort at consistent accuracy.

**Constructed tasks:** 13 in EfficientOpt.

Source: OptTips / T16.

## Technique cards

T17–T20

## FAMILY G

## T17 Dantzig–Wolfe decomposition and column generation

**Core idea:** Represent local feasible patterns as columns and generate improving ones by pricing instead of enumerating all patterns.

**Applicable patterns:** Routing, crew scheduling, cutting stock, and shifts; master normalization depends on the pattern interpretation.

**Inefficiency symptoms:** Pattern enumeration is prohibitive, while a fixed subset misses improving choices.

**Expert actions:** Start with a feasible restricted master or Phase I. Derive dual prices and solve pricing; use branch-and-price when integer certification requires it.

**Mathematical example:** Bounded local polytope  $X$  has vertices  $v_p$  and costs  $c_p = c^\top v_p$ . Retain the coupling requirement  $Ax \geq b$ .

## Compact linear formulation

$$\begin{array}{ll} \min & c^\top x \\ \text{s.t.} & Ax \geq b, \\ & x \in X \end{array}$$

## Full column master

$$\begin{array}{ll} \min & \sum_p c_p \lambda_p \\ \text{s.t.} & \sum_p A v_p \lambda_p \geq b, \\ & \sum_p \lambda_p = 1, \quad \lambda_p \geq 0 \end{array}$$

For  $\pi \geq 0$  and free  $\sigma$ , reduced cost is  $c_p - \pi^\top A v_p - \sigma$ . At an optimal restricted master, certified pricing with no negative reduced cost proves full-LP optimality.

**Benchmark guidance:** Validate patterns and reduced costs. Count master/pricing effort; certify absence of improving columns globally and verify required integer decisions separately.

**Constructed tasks:** 5 in EfficientOpt.

Source: OptTips / T17.

## FAMILY D

## T18 Network-flow and matching formulations

**Core idea:** Expose network or bipartite structure so integral extreme points permit continuous optimization or specialized network methods.

**Applicable patterns:** Transportation, transshipment, paths, and bipartite matching with integral data and compatible constraints.

**Inefficiency symptoms:** Path enumeration grows rapidly, or unnecessary integer declarations hide network structure. Side constraints can destroy integrality.

**Expert actions:** Verify the incidence structure and integral bounds. Obtain an integral optimal extreme point through a compatible method or recovery.

**Mathematical example:** Let  $B$  be a directed incidence matrix, with integral balanced supplies  $b$  and nonnegative integral finite capacities  $u$ .

## Integer network model

$$\begin{array}{ll} \min & c^\top x \\ Bx = b, & 0 \leq x \leq u \\ x \in \mathbb{Z}^{|A|} \end{array}$$

## Network LP

$$\begin{array}{ll} \min & c^\top x \\ Bx = b, & 0 \leq x \leq u \\ x \in \mathbb{R}^{|A|} \end{array}$$

Integral extreme points ensure an integral optimum, but an arbitrary optimum under ties may be fractional. Side constraints require rechecking; see T44 for transportation.

**Benchmark guidance:** Keep supplies, capacities, and costs fixed. Check returned decision integrality and feasibility; compare dimensions and total solve/recovery work.

**Selected case:** T18\_001 (p. 16)

OptTips / T18 · 21 EfficientOpt tasks.

## FAMILY D

## T19 Covering, partitioning, packing, and knapsack formulations

**Core idea:** Replace elementwise assignments by feasible patterns, then enforce covering, partitioning, or packing requirements explicitly.

**Applicable patterns:** Crew schedules, cutting stock, routes, and bin packing with well-defined pattern costs and coverage semantics.

**Inefficiency symptoms:** Assignment logic obscures compatibility; pattern enumeration explodes or the wrong coverage sense changes the task.

**Expert actions:** Define feasible patterns and reconstruction. Generate columns when useful; preserve the required integer selection.

**Mathematical example:** Bin packing:  $n$  items,  $0 < w_i \leq C$ , and  $n$  identical bins.  $P$  contains all nonempty subsets with total size at most  $C$ .

## Item–bin assignments

$$\begin{array}{ll} \min & \sum_b y_b \\ \sum_b x_{ib} = 1 & (\forall i) \\ \sum_i w_i x_{ib} \leq C y_b & (\forall b) \\ x_{ib}, y_b \in \{0, 1\} \end{array}$$

## Full pattern master

$$\begin{array}{ll} \min & \sum_{p \in P} \lambda_p \\ \sum_{p \ni i} \lambda_p = 1 & (\forall i) \\ \lambda_p \in \{0, 1\} \end{array}$$

Selected patterns reconstruct bins. Full integer models have the same optimum up to bin relabeling; restricting patterns can lose optimality.

**Benchmark guidance:** Check reconstructed assignments and costs. Certify pricing for LP bounds and integer optimality separately; count pattern-generation effort.

**Constructed tasks:** 14 in EfficientOpt.

Source: OptTips / T19.

## FAMILY D

## T20 Scheduling and temporal formulations

**Core idea:** Express start times and sequencing directly instead of enumerating job orders. Match the temporal representation to domains, resource rules, and objectives.

**Applicable patterns:** Single machine, parallel machine, job shop, workforce, and project scheduling require appropriate treatment of preemption, assignment, deadlines, and time domains.

**Inefficiency symptoms:** Permutation enumeration grows rapidly; unbounded starts weaken disjunctions. Interchangeable machines create symmetry, while an unnecessarily fine time grid enlarges formulations.

**Expert actions:** Bound the horizon, derive valid constants, and preserve precedence and capacities. Select disjunctive MILP, justified time indexing, or supported interval constraints.

**Mathematical example:** Two mandatory nonpreemptive jobs share one machine, with positive durations and bounded starts  $0 \leq s_k \leq H - p_k$ .

## Nonoverlap disjunction

$$\begin{array}{l} s_i + p_i \leq s_j \\ \text{or} \\ s_j + p_j \leq s_i \end{array}$$

## Bounded linear disjunction

$$\begin{array}{l} s_i + p_i \leq s_j + H(1 - z) \\ s_j + p_j \leq s_i + Hz \\ z \in \{0, 1\} \end{array}$$

These bounds make  $H$  valid in both inequalities. Optional jobs or multiple machines require additional activation or assignment logic.

**Benchmark guidance:** Keep durations, calendars, precedence, resources, and objectives fixed. Validate schedules and discretization, then compare representation and search under equivalent stopping criteria.

**Constructed tasks:** 43 in EfficientOpt.

Source: OptTips / T20.

## Technique cards

T21–T24

## FAMILY D

## T21 Routing and path formulations

**Core idea:** Combine degree equations with connectivity and resource restrictions to describe permitted journeys, not disconnected arc collections.

**Applicable patterns:** Traveling salesperson, capacitated routing, time windows, and network design require task-specific connectivity and resource conditions.

**Inefficiency symptoms:** Degree equations admit disconnected cycles; missing capacities or timing rules invalidate apparently cheap routes.

**Expert actions:** Use complete ordering, flow, subtour, or route-column formulations. Enforce necessary connectivity cuts throughout optimization.

**Mathematical example:** Directed TSP:  $n \geq 2$ ,  $V = \{1, \dots, n\}$ , no self arcs, binary  $x$ . Both minimize the same arc cost with  $\sum_{j \neq i} x_{ij} = \sum_{j \neq i} x_{ji} = 1$  for each  $i$ .

## MTZ ordering

$$\begin{aligned} 1 \leq u_i \leq n-1 \quad (i \neq 1) \\ u_i - u_j + (n-1)x_{ij} \leq n-2 \\ i, j \neq 1, \quad i \neq j \end{aligned}$$

## Single-commodity flow

$$\begin{aligned} 0 \leq f_{ij} \leq (n-1)x_{ij} \\ \sum_{j \neq i} (f_{ji} - f_{ij}) = 1 \\ i \neq 1 \end{aligned}$$

Each nondepot vertex consumes one unit; depot 1 supplies  $n-1$ . Either panel enforces connectivity with the shared degree model. Both  $f$  and  $u$  may be continuous.

**Benchmark guidance:** Compare complete formulations; verify each returned tour and resource condition. Include separation overhead and search work.

**Constructed tasks:** 21 in EfficientOpt.

Source: OptTips / T21.

## FAMILY H

## T22 Robust counterpart reformulation

**Core idea:** Replace uncertainty-indexed constraints with finite robust counterparts. Duality or conic structure preserves the same worst case guarantee without enumerating every disturbance.

**Applicable patterns:** Uncertain demand, resource coefficients, portfolio exposures, reserves, and regression errors admit explicit uncertainty sets; polyhedral, ellipsoidal, and budgeted sets differ structurally.

**Inefficiency symptoms:** Scenario enumeration can create excessive rows or miss unlisted outcomes. An unresolved worst case subproblem hides structure; altered uncertainty changes conservatism.

**Expert actions:** Write the support function explicitly. Use LP duality or suitable conic representations, checking assumptions that establish an exact robust counterpart.

**Mathematical example:** Uncertainty lies in the nonempty compact polyhedron  $U = \{u : Du \leq d\}$ ; the uncertain row coefficients are  $a_0 + Bu$ .

## Robust row

$$\begin{aligned} (a_0 + Bu)^T x \leq b \quad (\forall u \in U) \\ U = \{u : Du \leq d\} \end{aligned}$$

## Polyhedral counterpart

$$\begin{aligned} D^T \lambda = B^T x, \quad \lambda \geq 0 \\ a_0^T x + d^T \lambda \leq b \end{aligned}$$

An admissible  $\lambda$  certifies the robust row. Nonempty compact uncertainty ensures a finite inner LP optimum and its dual representation.

**Benchmark guidance:** Preserve uncertainty, decision timing, objective, and deterministic restrictions. Verify worst case feasibility and duality, then compare dimensions and solver work.

**Selected case:** [T22\\_008](#) (p. 17)

OptTips / T22 · 27 EfficientOpt tasks.

## FAMILY H

## T23 Stochastic programming and scenario decomposition

**Core idea:** Distinguish commitments before uncertainty from recourse afterward. Preserve nonanticipativity while exposing scenario structure for an extensive formulation or justified decomposition.

**Applicable patterns:** Inventory, supply chain, energy, capacity planning, and financial decisions combine shared commitments with scenario adjustments; probabilities and information timing must be explicit.

**Inefficiency symptoms:** Freely varying initial decisions across scenarios creates anticipation. Undifferentiated scenario expansion obscures separability, while deleting difficult scenarios changes the modeled risk.

**Expert actions:** Identify stages and probabilities, share initial variables or impose consensus, and isolate recourse. Choose decomposition guarantees appropriate to the variable domains.

**Mathematical example:** Probabilities  $p_s > 0$  sum to one;  $x \in X$  is shared across scenarios, with continuous linear recourse  $y_s \geq 0$ .

## Extensive formulation

$$\begin{aligned} \min \quad & c^T x + \sum_s p_s q_s^T y_s \\ \text{s.t.} \quad & W_s y_s \geq h_s - T_s x \quad (\forall s) \\ & x \in X, \quad y_s \geq 0 \end{aligned}$$

## Recourse value formulation

$$\begin{aligned} \min_{x \in X} \quad & c^T x + \sum_s p_s Q_s(x) \\ \text{s.t.} \quad & Q_s(x) = \min_{y_s \geq 0} \{q_s^T y_s : \\ & W_s y_s \geq h_s - T_s x\} \end{aligned}$$

Share one first-stage decision. Assume recourse is never unbounded below; assign  $Q_s(x) = +\infty$  when infeasible, excluding that decision.

**Benchmark guidance:** Preserve scenarios, probabilities, information, and objectives. Verify nonanticipativity and recourse feasibility, compare valid bounds, and distinguish certified methods from heuristic coordination.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T23.

## FAMILY B

## T24 Preprocessing and dominance elimination

**Core idea:** Remove impossible choices and proven redundant objects before model construction, preserving required outputs and the optimal value.

**Applicable patterns:** Assignments, schedules, routes, and patterns with incompatible or dominated choices.

**Inefficiency symptoms:** Dense index sets create immediately fixed variables. Dominated alternatives and redundant rows increase construction and presolve work.

**Expert actions:** Build sparse compatible index sets and reproducible filters. Prove every dominance replacement preserves remaining restrictions and does not worsen the objective.

**Mathematical example:** Let  $E$  contain compatible task–machine pairs. Other assignments are prohibited; costs, domains, and remaining constraints stay unchanged.

## Dense variables with fixings

$$\begin{aligned} x_{ij} &\geq 0 \quad ((i, j) \in I \times J) \\ x_{ij} &= 0 \quad ((i, j) \notin E) \end{aligned}$$

## Sparse indexing

$$\begin{aligned} x_{ij} &\geq 0 \quad ((i, j) \in E) \\ \hat{x}_{ij} &= \begin{cases} x_{ij}, & (i, j) \in E \\ 0, & (i, j) \notin E \end{cases} \end{aligned}$$

Substitute zero for removed variables everywhere. Set inclusion alone does not prove dominance for partitioning, packing, or additional resource restrictions.

**Benchmark guidance:** Preserve data and required outputs. Verify exclusions, reconstruct removed fixed values, and compare representation and solver work while accounting for preprocessing.

**Constructed tasks:** 18 in EfficientOpt.

Source: OptTips / T24.

## Technique cards

T25–T28

## FAMILY D

## T25 Symmetry breaking

**Core idea:** Select representatives of interchangeable objects to reduce repeated relabeling searches. Valid symmetry constraints retain an optimal representative without changing its objective.

**Applicable patterns:** Identical machines, vehicles, bins, colors, and clusters permit permutations only when costs, capacities, eligibility, and all constraints remain invariant under relabeling.

**Inefficiency symptoms:** Search revisits assignments differing only by labels. Arbitrary activation patterns multiply equivalent solutions; unjustified ordering may restrict objects that are not interchangeable.

**Expert actions:** Identify model-preserving permutations and impose justified usage, load, or lexicographic ordering. Prove every feasible orbit retains a representative and recover labels.

**Mathematical example:** Binary  $y_b$  indicates bin use. The feasible set  $X$  and objective  $f$  are invariant under every bin permutation.

## Interchangeable bins

$$\min_{(x,y) \in X, y \in \{0,1\}^B} f(x,y)$$

## Ordered representatives

$$\min_{(x,y) \in X, y \in \{0,1\}^B} f(x,y) \\ y_1 \geq y_2 \geq \dots \geq y_B$$

Relabel used bins before unused bins and permute assignments consistently. Heterogeneous or label-constrained bins require a different validity argument.

**Benchmark guidance:** Compare the same task up to authorized relabeling. Validate retained representatives and output requirements, then examine search work without claiming labeled-set equivalence.

**Constructed tasks:** 19 in EfficientOpt.

Source: OptTips / T25.

## FAMILY E

## T26 Complementarity and KKT/MPEC modeling

**Core idea:** Encode complementary activity completely. Replacing lower optimization by KKT conditions additionally requires assumptions establishing that those conditions characterize the intended lower solutions.

**Applicable patterns:** Equilibrium, bilevel, active-constraint, and MPEC models contain complementary pairs. KKT reformulations require attention to differentiability, convexity, constraint qualifications, and multiplier existence.

**Inefficiency symptoms:** Product equalities can omit sign restrictions. Missing dual feasibility or stationarity, and unjustified KKT substitution for nonconvex problems, change the task.

**Expert actions:** Check KKT validity and include all primal, dual, stationarity, and complementarity conditions. Use supported SOS1, valid bounded disjunctions, or suitable MPEC methods.

**Mathematical example:** For one complementary pair,  $a, b$  are nonnegative continuous quantities with proven bounds  $0 \leq U, V < \infty$ .

## Bounded complementarity

$$0 \leq a \leq U, \quad 0 \leq b \leq V \\ ab = 0$$

## Binary disjunction

$$0 \leq a \leq Uz \\ 0 \leq b \leq V(1-z) \\ z \in \{0,1\}$$

The bounded disjunction is exact. Correct complementarity encoding alone does not establish that KKT conditions characterize global lower-level optima.

**Benchmark guidance:** Compare product and disjunctive encodings with identical bounds; verify complementarity and required outputs, and measure runtime under matched global optimality tolerances.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T26.

## FAMILY H

## T27 Multi-objective and lexicographic optimization

**Core idea:** Express objective priorities through sequential optimization and explicit bounds. Arbitrarily large weights can distort scaling without enforcing the intended hierarchy.

**Applicable patterns:** Service, scheduling, and risk–return priorities. Distinguish strict lexicographic order, weighted preferences, and permitted degradation of earlier objectives.

**Inefficiency symptoms:** Huge coefficients distort objective scaling yet may fail to enforce priorities. Unjustified weights let secondary improvements compromise a supposedly dominant objective.

**Expert actions:** Define priorities and tolerances, optimize sequentially or hierarchically, and retain earlier objective bounds. Use normalized weights only for genuinely weighted preferences.

**Mathematical example:** Both objectives are minimized on  $X$ ; the attained first minimum has value  $z_1^*$ . The second stage preserves its specified tolerance.

## First priority

$$z_1^* = \min_{x \in X} f_1(x)$$

## Second priority

$$\min_{x \in X} f_2(x) \\ f_1(x) \leq z_1^* + \varepsilon \\ \varepsilon \geq 0$$

Zero tolerance gives the exact hierarchy. Positive tolerance permits declared first-objective degradation and must be part of the task specification.

**Benchmark guidance:** Certify the first-stage optimum before fixing its bound; compare every objective level and total multistage cost under identical degradation tolerances.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T27.

## FAMILY H

## T28 Penalty, slack, and feasibility-relaxation modeling

**Core idea:** Introduce interpretable slacks and calibrated penalties only for authorized soft requirements. Separate feasibility diagnosis from final optimization and report every relaxed requirement.

**Applicable patterns:** Demand shortfalls, service levels, shortages, inconsistent data, and contractual violations may permit relaxation; mandatory restrictions and business penalties must remain explicit.

**Inefficiency symptoms:** Unexplained infeasibility gives little guidance. Arbitrary penalties distort scaling, and silently relaxing mandatory requirements produces a solution to a different problem.

**Expert actions:** Classify hard and soft rows, define meaningful slack units, and justify penalties or priorities. Separate restoration when needed and report final violations.

**Mathematical example:** Let  $X$  contain mandatory restrictions. Demand  $d$  permits nonnegative shortfall  $s$ , explicitly penalized at rate  $\rho > 0$ .

## Hard requirement

$$\min_{x \in X} f(x) \\ a^T x \geq d$$

## Authorized soft requirement

$$\min_{x \in X, s \geq 0} f(x) + \rho s \\ a^T x + s \geq d$$

The soft model changes the specification. Finite penalties do not automatically enforce hard feasibility; distinguish restoration from final optimization.

**Benchmark guidance:** Compare implementations of the same authorized relaxation; report original constraint residuals, individual slacks, penalty contributions, and total runtime at matched tolerances.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T28.

## Technique cards

T29–T32

## FAMILY H

## T29 Solver-native modeling constructs

**Core idea:** Express recognized structure through supported native constraints. Compatible solvers can interpret mathematical relations directly instead of requiring every disjunction or relation expanded manually.

**Applicable patterns:** Indicators, SOS, piecewise-linear relations, intervals, cumulative resources, and cones have specialized representations. Suitability depends on support, domains, optionality, and exact semantics.

**Inefficiency symptoms:** Handwritten expansions become lengthy and fragile. Incorrect segment adjacency, missing activation logic, or unsuitable scheduling representations hide structure and can change feasibility.

**Expert actions:** Choose native constructs with matching semantics and document assumptions. Retain a valid alternative encoding and inspect interface support and export transformations.

**Mathematical example:** Mandatory intervals  $I_i = [s_i, e_i)$  share one unary resource, with fixed positive durations and  $e_i = s_i + p_i$ .

## Pairwise disjunctions

$$e_i = s_i + p_i \quad (\forall i) \\ e_i \leq s_j \vee e_j \leq s_i \quad (i < j)$$

## Native interval relation

$$I_i = [s_i, s_i + p_i) \quad (\forall i) \\ \text{NoOverlap}(I_1, \dots, I_n)$$

Both descriptions enforce mandatory nonoverlap. Integer-time interfaces need exact domain compatibility or justified discretization; optional intervals require matching presence semantics.

**Benchmark guidance:** Implement pairwise and native nonoverlap with identical start domains, durations, and presence; validate schedules and compare construction time, search time, and memory.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T29.

## FAMILY H

## T30 Warm starts and incumbent injection

**Core idea:** Provide validated initial solutions or supported partial hints to guide search. Starts supply incumbent information without changing the model or fixing decisions.

**Applicable patterns:** Mixed integer planning, routing, scheduling, facilities, and networks can use greedy, historical, repaired, or rounded candidates that respect the current instance.

**Inefficiency symptoms:** Search lacks useful incumbents despite available knowledge. Historical starts may violate changed data, while blind rounding breaks coupling constraints or partial assignments.

**Expert actions:** Construct, validate, and appropriately repair candidates, then submit supported starts or hints. Record whether each complete candidate is feasible and accepted.

**Mathematical example:** For minimization over  $X$ , a complete validated candidate  $x^0 \in X$  supplies the objective upper bound  $U = f(x^0)$ .

## Unchanged optimization task

$$f^* = \min_{x \in X} f(x)$$

## Validated incumbent information

$$x^0 \in X, \quad U = f(x^0) \\ x_i^{\text{start}} = x_i^0 \\ L_{\text{node}} \geq U \Rightarrow \text{prune}$$

Pruning requires valid node lower bounds. Partial hints are not feasible incumbents, and starts must not become unintended variable fixings.

**Benchmark guidance:** Preserve models, instances, stopping criteria, and outputs. Account for start construction and validation, compare bound progress, and distinguish feasible incumbents from hints.

**Constructed tasks:** 16 in EfficientOpt.

Source: OptTips / T30.

## FAMILY H

## T31 Algorithm–structure matching

**Core idea:** Choose algorithms whose assumptions and guarantees match the model's structure, domains, and required outputs.

**Applicable patterns:** Models with exploitable linear, conic, network, or discrete structure. Convexity, sparsity, and variable domains determine which methods apply.

**Inefficiency symptoms:** Generic nonlinear interfaces can hide convex cones, while unnecessary integrality conceals networks. Unsupported domains or mismatched solution guarantees invalidate nominal comparisons.

**Expert actions:** Classify structure, verify assumptions, and choose compatible formulations and algorithms. Preserve outputs and distinguish feasibility, local stationarity, and global optimality guarantees.

**Mathematical example:** For continuous  $x$ , consider an affine norm constraint. The remaining set  $X$  is convex and admits a compatible conic representation.

## Affine norm constraint

$$\min_{x \in X} c^T x \\ \|Ax + b\|_2 \leq d^T x + e$$

## Second-order cone form

$$\min_{x \in X} c^T x \\ (d^T x + e, Ax + b) \in Q \\ Q = \{(t, v) : \|v\|_2 \leq t\}$$

Cone membership preserves the nonnegative right-hand side. Integer restrictions instead require a mixed integer conic method with matching guarantees.

**Benchmark guidance:** Compare the same task with matched correctness and stopping requirements. Document interfaces and reformulations, validate recovery, and distinguish local from certified optima.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T31.

## FAMILY H

## T32 Solver-log-guided diagnosis

**Core idea:** Use logs as evidence about bounds, search, scaling, and model size. Form testable reformulation hypotheses rather than interpreting stopping status as explanation.

**Applicable patterns:** Model reviews and diagnostic loops inspect root bounds, incumbents, nodes, coefficient ranges, presolve, and cuts when the solver exposes relevant information.

**Inefficiency symptoms:** Termination status alone conceals difficulties with feasibility, bounds, numerics, or size. Repeated code edits without structural evidence can leave modeling issues unresolved.

**Expert actions:** Extract documented quantities and definitions. Relate observed bounds, search, and scaling to plausible causes, then validate targeted changes without assuming causality.

**Mathematical example:** For minimization, finite  $L, U$  bound the optimum below and above. Matrix  $A$  contains the recorded constraint coefficients.

## Bound progress summary

$$L \leq f^* \leq U \\ g_{\text{abs}} = U - L$$

## Coefficient spread diagnostic

$$\mathcal{A} = \{|A_{ij}| : A_{ij} \neq 0\} \\ R_A = \frac{\max \mathcal{A}}{\min \mathcal{A}} \\ \mathcal{A} \neq \emptyset$$

Gap and coefficient spread diagnose different issues. Neither a large ratio nor a large search tree uniquely identifies its cause.

**Benchmark guidance:** Change one modeling choice at a time; keep data and settings fixed, validate correctness, and compare root bounds, search progress, numerical warnings, and runtime.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T32.

## Technique cards

T33–T36

## FAMILY A

## T33 Standard/canonical-form transformation

**Core idea:** Organize objectives, constraints, and domains into canonical forms, preserving the task while making solver mapping, verification, and code generation explicit.

**Applicable patterns:** Linear, quadratic, conic, and mixed integer models require consistent signs, bounds, domains, objective direction, and constants when mapped to solver interfaces.

**Inefficiency symptoms:** Mixed conventions obscure equivalence; incorrect conversions can reverse inequalities, omit domains, or silently change objective constants and reported values.

**Expert actions:** Classify the model, record all domains, and track substitutions and objective conversions; standardize the whole representation, including any necessary auxiliaries.

**Mathematical example:** Here  $x \in \mathbb{R}^n$  is unrestricted and  $Gx \leq h$  contains the original inequalities; each row receives a separate slack.

## Original LP

$$\min_{x \in \mathbb{R}^n} c^\top x \\ Gx \leq h$$

## Nonnegative equality form

$$\min c^\top (x^+ - x^-) \\ G(x^+ - x^-) + s = h \\ x^+, x^-, s \geq 0$$

Recover  $x = x^+ - x^-$ . This nonunique splitting preserves feasible decisions and objective values; required integer domains need separate bookkeeping.

**Benchmark guidance:** Build original and standard-form LPs; recover decisions, compare feasibility and objective values including constants, and measure construction and solve time.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T33.

## FAMILY A

## T35 Variable elimination and dimension reduction

**Core idea:** Use equalities or conservation laws to eliminate dependent variables, reducing dimension when substitution preserves useful sparsity and all original restrictions.

**Applicable patterns:** Consider simple affine equalities, conservation relations, and decisions confined to affine subspaces, with explicit recovery of eliminated variables without losing feasible solutions.

**Inefficiency symptoms:** Redundant variables and equalities enlarge the representation, burden presolve, or impair conditioning; excessive substitution can instead create dense expressions.

**Expert actions:** Substitute dependent variables throughout the objective and constraints, transfer bounds and domains, and retain variables when elimination would destroy sparsity.

**Mathematical example:** Let  $x, y \geq 0$  satisfy  $x + y = 10$ , with objective  $f$  defined over the resulting one dimensional feasible segment.

## Explicit equality

$$\min_{x,y} f(x, y) \\ x + y = 10 \\ x \geq 0, \quad y \geq 0$$

## Eliminated variable

$$\min_x f(x, 10 - x) \\ 0 \leq x \leq 10 \\ y = 10 - x$$

The recovery map preserves feasible decisions and objective values. For integral original variables, retain the induced integrality restrictions.

**Benchmark guidance:** Compare full and reduced representations on identical data, documenting eliminated variables, transferred restrictions, and recovery of requested decisions without assuming universal improvement.

**Constructed tasks:** 13 in EfficientOpt.

Source: OptTips / T35.

## FAMILY A

## T34 Slack, surplus, and artificial variables

**Core idea:** Nonnegative slack and surplus variables convert inequalities exactly into equalities. Artificial variables serve initialization or diagnosis, distinct from authorized soft-constraint violations.

**Applicable patterns:** LP standardization, simplex initialization, and restoration use different auxiliaries: slack records unused capacity, surplus records excess, and artificial variables measure diagnostic residuals.

**Inefficiency symptoms:** Missing slack signs weaken inequalities. Artificial variables can contaminate final objectives, while one-sided residuals may fail to represent both directions of equality violation.

**Expert actions:** Use correct slack signs and domains. Solve a suitable Phase I, inspect residuals, then remove or fix artificial variables before restoring the objective.

**Mathematical example:** Each inequality uses its own slack. Phase I separately tests  $Ax = b$ ,  $x \geq 0$ , using nonnegative artificial residuals.

## Exact slack and surplus

$$a^T x \leq b \iff a^T x + s = b \\ a^T x \geq b \iff a^T x - s = b \\ s \geq 0$$

## Equality Phase I

$$\min \mathbf{1}^T (r^+ + r^-) \\ Ax + r^+ - r^- = b \\ x, r^+, r^- \geq 0$$

Zero Phase-I optimum certifies feasibility; positive optimum diagnoses infeasibility in exact arithmetic. Two residual directions cover arbitrary equality violations.

The equality-phase example shown is  $L_1$  feasibility restoration using paired residual variables  $r^+, r^-$ .

**Benchmark guidance:** Validate standardization separately from restoration. Preserve original constraints and objective after initialization, report residuals, and do not present relaxed tasks as equivalent originals.

**Constructed tasks:** 16 in EfficientOpt.

Source: OptTips / T34.

## FAMILY F

## T36 Convexity recognition and DCP modeling

**Core idea:** Recognize convexity before choosing a representation; DCP rules and conic forms can expose structure hidden by a generic nonlinear expression.

**Applicable patterns:** Look for convex quadratic losses, norm bounds, risk constraints, and regularization; check objective direction, inequality orientation, affine equalities, and domains.

**Inefficiency symptoms:** Opaque expressions may hide convexity or fail DCP parsing; incorrect objective directions or inequality orientations instead change the mathematical problem.

**Expert actions:** Check composition rules or appropriate Hessian conditions, then introduce exact epigraphs or conic forms while preserving the objective, feasible set, and domains.

**Mathematical example:** For real  $A, b$ , least squares is convex; its squared norm objective has the exact rotated cone epigraph shown below.

## Convex quadratic

$$\min_x \|Ax - b\|_2^2$$

## Conic epigraph

$$\min_{x,t} t \\ (t, \tfrac{1}{2}, Ax - b) \in Q_r \\ Q_r = \{(u, v, z) : u, v \geq 0, \\ 2uv \geq \|z\|_2^2\}$$

DCP acceptance is sufficient, not necessary, for convexity. The Hessian criterion requires twice differentiability on a convex domain.

**Benchmark guidance:** Compare quadratic-objective and rotated-cone models on identical data; verify least-squares residuals and objective values, then measure construction time, solve time, and memory.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T36.

## Technique cards

T37–T40

## FAMILY F

## T37 Smooth–nonsmooth composite modeling

**Core idea:** Separate a smooth loss from a proximal nonsmooth regularizer, preserving the objective while exposing computational structure without assuming differentiability everywhere.

**Applicable patterns:** Consider Lasso, regularized logistic regression, total variation, and sparse or low rank learning; identify smooth and tractable proximal components explicitly.

**Inefficiency symptoms:** Ordinary derivatives fail at nonsmooth points, while unnecessary smoothing changes the objective and introduces an additional approximation parameter to manage.

**Expert actions:** Separate loss and regularizer; specify gradient regularity, proximal computability, and block structure, using an exact epigraph or compatible composite formulation.

**Mathematical example:** For  $\lambda \geq 0$ , Lasso combines a convex quadratic loss with an  $\ell_1$  penalty admitting a componentwise proximal operator.

Explicit  $\ell_1$  epigraph

$$\min_{x,t} \frac{1}{2} \|Ax - b\|_2^2 + \lambda \sum_i t_i \\ -t_i \leq x_i \leq t_i \quad \forall i$$

## Composite structure

$$\min_x f(x) + g(x) \\ f(x) = \frac{1}{2} \|Ax - b\|_2^2 \\ g(x) = \lambda \|x\|_1$$

Both forms have equal optimal values; for  $\lambda > 0$ , optimal epigraph variables equal absolute values. Composite structure alone provides no convergence guarantee.

**Benchmark guidance:** Compare a valid epigraph with an explicit composite representation of the same original objective; label smoothing as approximation and exclude invalid derivatives.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T37.

## FAMILY F

## T38 Convex conjugacy and Fenchel duality

**Core idea:** Use conjugates and support functions to express nested maximizations or derive dual problems; replacing a primal requires justified duality conditions.

**Applicable patterns:** Look for robust losses, regularization penalties, support functions, or nested linear maximization; identify functions and domains before eliminating an inner problem.

**Inefficiency symptoms:** Explicit supremums or scenario enumeration can obscure known conjugates; an unjustified dual substitution may change the attainable value or required outputs.

**Expert actions:** Derive conjugates, signs, and linear maps carefully; check duality qualifications and specify primal decision recovery when the task requires those decisions.

**Mathematical example:** Let  $f, g$  be proper closed convex functions and  $A$  linear; assume  $f$  continuous at  $A\bar{x}$  for some  $\bar{x} \in \text{dom } g$ .

## Primal and conjugate

$$\inf_x f(Ax) + g(x) \\ f^*(y) = \sup_z \{y^\top z - f(z)\}$$

## Fenchel dual

$$\sup_y -f^*(y) - g^*(-A^\top y)$$

With this qualification and finite primal optimal value, values agree. Feasible sets differ, and requested primal decisions require separate recovery.

**Benchmark guidance:** On the same instance, check primal and dual feasibility, the primal–dual gap, and recovered requested decisions; include recovery costs in timing.

**Constructed tasks:** 7 in EfficientOpt.

Source: OptTips / T38.

## FAMILY F

## T39 Subgradient, normal-cone, and optimality-condition modeling

**Core idea:** Use subdifferentials and normal cones for nonsmooth optimality; separating their contributions requires a valid sum rule rather than ordinary gradient equations.

**Applicable patterns:** Consider convex  $\ell_1$  losses, sparse regularization, and projection problems over convex sets; identify kinks, feasible domains, and the relevant sum rule qualification.

**Inefficiency symptoms:** Ordinary derivatives fail at absolute value or maximum kinks; omitted subgradients or normal cones can exclude optima or misrepresent constrained optimality.

**Expert actions:** Write the subdifferential inclusion with a consistent normal cone sign, verifying convexity and qualification before splitting terms into separate contributions.

**Mathematical example:** Let  $C$  be nonempty closed convex and  $f$  proper closed convex, continuous at some point of  $C$  where  $f$  is finite.

## Constrained minimization

$$\min_{x \in C} f(x) \\ N_C(x) = \{v : v^\top(y - x) \leq 0 \\ \forall y \in C\}, \quad x \in C$$

## Qualified optimality condition

$$x^* \in C \\ 0 \in \partial f(x^*) + N_C(x^*)$$

These assumptions make the inclusion characterize global minimizers. Without qualification, retain the unsplit subdifferential of  $f$  plus the indicator of  $C$ .

**Benchmark guidance:** Compare the original constrained problem with justified nonsmooth optimality conditions; incorrect derivatives are diagnostic failure cases, never valid ordinary formulations.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T39.

## FAMILY F

## T40 Projection and proximal-operator modeling

**Core idea:** Recognize dedicated projection and proximal operators before repeatedly invoking generic optimizers, exploiting closed forms or separability without changing the subproblem.

**Applicable patterns:** Inspect proximal gradient, projected gradient, or ADMM subproblems for simple convex projections and penalties with componentwise or spectral proximal mappings.

**Inefficiency symptoms:** Generic subproblem solves can overlook a closed form; incorrect scaling, domains, norms, or regularization parameters can invalidate an apparent shortcut.

**Expert actions:** Derive the operator and its scaling, check existence and uniqueness, and implement justified separable blocks without assuming every regularizer has a closed form.

**Mathematical example:** For real vector  $v$  and  $\lambda > 0$ , the  $\ell_1$  proximal objective is strongly convex, yielding a unique componentwise solution.

## Proximal subproblem

$$\min_x \lambda \|x\|_1 + \frac{1}{2} \|x - v\|_2^2$$

## Componentwise soft thresholding

$$x_i^* = \text{sign}(v_i) \\ \cdot \max\{|v_i| - \lambda, 0\}$$

The displayed scaling gives threshold  $\lambda$ . Projection onto a nonempty closed convex set is the proximal map of its indicator.

**Benchmark guidance:** Compare generic solves with soft thresholding for identical  $v, \lambda$ , including  $|v_i| = \lambda$ ; check objectives and optimality residuals, and measure per-call time.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T40.

## Technique cards

T41–T44

## FAMILY C

## T41 Variational-inequality and equilibrium-constraint modeling

**Core idea:** Preserve equilibrium through VI or complementarity conditions; agents optimizing separate objectives generally cannot be replaced by an arbitrary social objective.

**Applicable patterns:** Consider traffic, market, energy, or supply chain equilibria; specify the feasible set and response mapping before choosing a compatible equilibrium representation.

**Inefficiency symptoms:** A social objective can describe a different problem without a justified potential; missing feasibility or complementarity conditions also compromise equilibrium modeling.

**Expert actions:** Define the feasible set and mapping, inspect monotonicity and complementarity, and choose a compatible representation and solver; justify potential models separately.

**Mathematical example:** The complementarity form below applies specifically to  $K = \mathbb{R}_+^n$ ; a VI on a general feasible set need not admit it.

## Variational inequality

$$\begin{aligned} \text{find } x^* \in \mathbb{R}_+^n \\ F(x^*)^\top (y - x^*) \geq 0 \\ \forall y \in \mathbb{R}_+^n \end{aligned}$$

## Nonlinear complementarity

$$\begin{aligned} x^* \geq 0, \quad F(x^*) \geq 0 \\ x_i^* F_i(x^*) = 0 \quad \forall i \end{aligned}$$

These conditions are equivalent on the nonnegative orthant. Monotonicity alone neither supplies a scalar potential nor guarantees uniqueness.

**Benchmark guidance:** Compare VI and complementarity formulations using the same  $F$  and domain; check nonnegativity and complementarity residuals, and compare runtime at matched tolerances.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T41.

## FAMILY C

## T42 Sensitivity and parametric optimization

**Core idea:** Expose changing coefficients as parameters for reproducible instance comparisons and sensitivity analysis; dual values provide information subject to appropriate validity conditions.

**Applicable patterns:** Consider transportation, resource allocation, capacity expansion, and pricing under perturbations; distinguish changing coefficients, structural parameters, decisions, and reported solution quantities.

**Inefficiency symptoms:** Hardcoded coefficients obstruct controlled comparisons; shadow prices are ambiguous without the associated constraint, inequality direction, sign convention, and validity range.

**Expert actions:** Declare parameters, record active constraints and dual values, and test perturbations; avoid extrapolating local value predictions beyond their valid region.

**Mathematical example:** Assume finite primal and dual optima at  $b$ ; a dual optimum  $y^*$  supplies a supporting bound for perturbed LP values.

## Parameterized primal

$$\begin{aligned} v(b) = \min_x \quad c^\top x \\ Ax \geq b, \quad x \geq 0 \end{aligned}$$

## Dual and sensitivity bound

$$\begin{aligned} v(b) = \max_y \quad b^\top y \\ A^\top y \leq c, \quad y \geq 0 \\ v(b + \Delta b) \geq v(b) + (y^*)^\top \Delta b \end{aligned}$$

For feasible finite perturbed values, the bound holds; equality requires  $y^*$  to remain dual optimal. Kinks need not admit unique derivatives.

**Benchmark guidance:** Perturb selected right-hand sides and resolve each LP; compare actual value changes with dual supporting bounds, recording active-set changes and solve time.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T42.

## FAMILY G

## T43 Dynamic programming and Bellman recursion

**Core idea:** Represent repeated multistage subproblems through a sufficient state and Bellman recursion, preserving future consequences without automatically expanding every stage.

**Applicable patterns:** Finite-horizon inventory, equipment replacement, production, and resource allocation admit state, action, transition, and cost descriptions.

**Inefficiency symptoms:** Full expansion repeats subproblems; oversized states impede computation, while insufficient states omit dependencies and invalidate recursion.

**Expert actions:** Specify states, admissible actions, transitions, and terminal costs; verify state sufficiency before implementing recursive or hybrid solution methods.

**Mathematical example:** Assume deterministic transitions, a fixed initial state, and a state retaining all information needed for future decisions.

## Full horizon decisions

$$\begin{aligned} \min \quad \sum_{t=0}^{T-1} c_t(s_t, a_t) + g(s_T) \\ s_{t+1} = F_t(s_t, a_t) \\ a_t \in \mathcal{A}_t(s_t), \quad s_0 = \bar{s} \end{aligned}$$

## Bellman recursion

$$\begin{aligned} V_T(s) &= g(s) \\ V_t(s) &= \min_{a \in \mathcal{A}_t(s)} \{c_t(s, a) + V_{t+1}(F_t(s, a))\} \end{aligned}$$

With attained stage minima and sufficient states, both forms give the same objective; recursion still faces dimensionality limits.

**Benchmark guidance:** Compare full-horizon and Bellman implementations with identical transitions and costs; recover actions, verify objectives, and measure memory and runtime.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T43.

## FAMILY D

## T44 Transportation and network-simplex structure

**Core idea:** Use transportation and transshipment network structure to remove explicit integrality when integral-data assumptions justify it.

**Applicable patterns:** Look for linear cost shipments between supply, demand, or transshipment nodes; identify balance equations, capacities, and side constraints affecting network structure.

**Inefficiency symptoms:** Generic integer formulations may obscure pure transportation structure, but logical decisions or coupling constraints can invalidate network arguments and cannot be ignored.

**Expert actions:** Identify nodes, arcs, balances, and costs. Preserve incidence structure and justify integrality relaxation before choosing a network method.

**Mathematical example:** Let supplies and demands be nonnegative integers with equal totals, and assume a feasible transportation network with linear costs.

## Integer transportation

$$\begin{aligned} \min_x \quad \sum_{i,j} c_{ij} x_{ij} \\ \sum_j x_{ij} = s_i \quad \forall i \\ \sum_i x_{ij} = d_j \quad \forall j \\ x_{ij} \in \mathbb{Z}_{\geq 0} \end{aligned}$$

## Network LP

$$\begin{aligned} \min_x \quad \sum_{i,j} c_{ij} x_{ij} \\ \sum_j x_{ij} = s_i \quad \forall i \\ \sum_i x_{ij} = d_j \quad \forall j \\ x_{ij} \geq 0 \end{aligned}$$

Total unimodularity ensures integral extreme points and an integral optimum; arbitrary optimal LP solutions may be fractional when optima are nonunique.

T18 concerns discovering or constructing network-flow/matching formulations for broader combinatorial problems; T44 focuses on preserving an already-present transportation/transshipment structure for specialized network optimization.

**Benchmark guidance:** Compare integer and network LP models on identical instances; obtain a basic or network optimum, verify integral flows, and report objective, time, and memory.

**Constructed tasks:** 14 in EfficientOpt.

Source: OptTips / T44.

# Technique cards

T45–T48

## FAMILY C

### T45 Constraint qualifications and KKT validity

**Core idea:** Convexity supports KKT sufficiency; constraint qualifications justify necessity and multiplier existence when replacing optimization by optimality conditions.

**Applicable patterns:** Nonlinear, bilevel, equilibrium, and dual models require explicit differentiability, convexity, domains, and qualification checks.

**Inefficiency symptoms:** Unjustified KKT substitution can admit nonoptimal lower-level points or exclude genuine optima lacking multipliers.

**Expert actions:** Distinguish Slater, LICQ, and MFCQ assumptions and roles; prove equivalence or retain the original optimization problem.

**Mathematical example:** Let  $f, g_i$  be finite differentiable convex functions on  $\mathbb{R}^n$ , with an attained finite minimum, Slater feasibility, affine equalities, and free  $v$ .

#### Convex primal

$$\begin{aligned} \min_x \quad & f(x) \\ g_i(x) & \leq 0 \quad \forall i \\ Ax & = b \end{aligned}$$

#### KKT characterization

$$\begin{aligned} \nabla f(x) + A^\top v \\ + \sum_i \mu_i \nabla g_i(x) & = 0 \\ g_i(x) & \leq 0, \quad Ax = b \\ \mu_i & \geq 0, \quad \mu_i g_i(x) = 0 \quad (\forall i) \end{aligned}$$

Slater requires strict inequalities and affine feasibility; it is sufficient, not necessary. Nonconvex KKT points need not be globally optimal. T26 encodes complementarity/KKT systems computationally; T45 establishes whether KKT conditions validly characterize the underlying optimization problem. Benchmark guidance: Compare equivalent primal and KKT models; verify feasibility, stationarity, complementarity, objective agreement, and requested decisions before comparing runtime.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T45.

## FAMILY D

### T47 Matrix-variable and low-rank modeling

**Core idea:** Preserve matrix decisions and spectral structure; nuclear norm surrogates and rank factorizations have different roles and are not automatically equivalent reformulations.

**Applicable patterns:** Consider matrix completion, robust PCA, covariance estimation, low rank recovery, and SDP; identify dimensions, observations, rank assumptions, and relevant spectral restrictions.

**Inefficiency symptoms:** Vectorization can obscure matrix operators, while direct rank terms remain nonconvex; unstated surrogate or factorization assumptions can misrepresent the intended task.

**Expert actions:** Retain matrix operators, select justified PSD, nuclear norm, or factorized representations, and state rank and convexity assumptions separately from objective changes.

**Mathematical example:** For real matrices  $X, M$ ,  $P_\Omega$  retains observed entries and zeros others;  $\|X\|_* = \sum_i \sigma_i(X)$  penalizes singular values, with  $\lambda > 0$ .

#### Rank penalization

$$\min_X \quad \frac{1}{2} \|P_\Omega(X - M)\|_F^2 + \lambda \text{rank}(X)$$

#### Convex nuclear norm surrogate

$$\min_X \quad \frac{1}{2} \|P_\Omega(X - M)\|_F^2 + \lambda \|X\|_*$$

The surrogate need not share the optimum. Factorization  $X = UV^\top$  with  $r$  columns restricts rank to at most  $r$  and is nonconvex.

**Benchmark guidance:** Use identical observation and validation masks; compare reconstruction error, achieved rank, memory, and runtime, reporting each penalty and tuning rule separately.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T47.

## FAMILY F

### T46 Interior-point-friendly modeling

**Core idea:** Preserve sparse operators, compatible scales, and standard convex structure so interior point linear algebra can exploit the model rather than dense expansions.

**Applicable patterns:** Continuous convex LP, QP, SOCP, SDP, and mixed conic models benefit from inspecting operator sparsity, compatible cones, and scaling.

**Inefficiency symptoms:** Forming dense Gram matrices, oversized penalties, or poorly scaled constraints can increase factorization work and aggravate numerical difficulties.

**Expert actions:** Preserve sparse operators through residual variables or standard cones, scale consistently, and inspect presolve and factorization behavior before selecting the representation.

**Mathematical example:** For sparse  $A$ , fixed  $b$ , and the same convex polyhedral set  $X$ , the Gram matrix  $A^\top A$  can be dense.

#### Expanded Gram matrix

$$\begin{aligned} \min_{x \in X} \quad & \frac{1}{2} x^\top Q x - c^\top x + d \\ Q & = A^\top A, \quad c = A^\top b \\ d & = \frac{1}{2} b^\top b \end{aligned}$$

#### Sparse residual formulation

$$\begin{aligned} \min_{x \in X, r} \quad & \frac{1}{2} r^\top r \\ r & = Ax - b \end{aligned}$$

Substitution gives identical objectives. Retaining  $A$  avoids explicitly forming  $A^\top A$ ; presolve, ordering, and solver factorization still determine computational cost. T36 recognizes convexity and DCP-compatible structure; it does not primarily choose a cone representation, an algorithm, or an IPM-oriented numerical formulation. Benchmark guidance: Compare Gram-matrix and residual models with identical scaling and tolerances; record matrix nonzeros, available factorization time, peak memory, and total time.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T46.

## FAMILY B

### T48 Cover, clique, subtour, and capacity cuts

**Core idea:** Use problem specific valid inequalities or required feasibility constraints, recognizing distinct validity conditions for cover, clique, subtour, and capacity families.

**Applicable patterns:** Look for knapsack covers, exclusive choices, disconnected routing cycles, or vehicle capacity violations; identify combinatorial structure and variable conventions first.

**Inefficiency symptoms:** Weak relaxations admit fractional conflicts or capacity patterns; degree constraints alone can additionally admit disconnected integer cycles, which are invalid tours.

**Expert actions:** Derive valid inequalities under appropriate assumptions; use reviewed separation or lazy enforcement for large families, distinguishing correctness constraints from optional strengthening.

**Mathematical example:** Use binary arcs  $x_{ij}$  for  $i \neq j$ , unit incoming and outgoing degrees, and the same tour-cost objective, with  $n = |V| \geq 3$ .

#### Complete MTZ formulation

$$\begin{aligned} u_i - u_j + nx_{ij} & \leq n - 1 \\ i & \neq j, \quad i, j \in V \setminus \{1\} \\ 1 & \leq u_i \leq n - 1 \quad (i \neq 1) \end{aligned}$$

#### Subtour elimination constraints

$$\begin{aligned} \sum_{i \in S} \sum_{j \in S \setminus \{i\}} x_{ij} \\ & \leq |S| - 1 \\ S & \subset V, \quad 2 \leq |S| < n \end{aligned}$$

Continuous ordering variables suffice. SECs must exclude every disconnected integer candidate; optional relaxation cuts alone do not guarantee tour feasibility. T07 addresses the general methodology of valid inequalities, separation, user cuts, and lazy constraints; T48 identifies specific combinatorial cut families. Benchmark guidance: Compare MTZ with complete SEC enforcement on identical directed instances; verify a single Hamiltonian tour and compare root bounds, nodes, and total time.

**Constructed tasks:** 15 in EfficientOpt.

Source: OptTips / T48.

## Technique cards

T49–T50

## FAMILY B

## T49 Linking constraints and activation strengthening

**Core idea:** Add direct activation links to strengthen relaxations, even when aggregated capacity activation already enforces the relationship at feasible integer decisions.

**Applicable patterns:** Consider facility location, fixed charge production, activated flows, and switchable resources; identify decisions requiring activation and whether links preserve integer feasibility.

**Inefficiency symptoms:** Aggregated capacity activation can let a fractional opening support a large individual assignment, leaving weak fractional patterns without direct disaggregated links.

**Expert actions:** Add valid decision to activation links while retaining capacity restrictions; tighten justified bounds and inspect fractional openings without changing integer feasibility.

**Mathematical example:** For facility  $j$ , assume  $d_i > 0$  and  $C_j > 0$ , with binary assignments  $x_{ij}$  and opening decision  $y_j$ .

## Aggregated activation

$$\sum_i d_i x_{ij} \leq C_j y_j \\ x_{ij} \in \{0, 1\}, \quad y_j \in \{0, 1\}$$

## Disaggregated strengthening

$$\sum_i d_i x_{ij} \leq C_j y_j \\ x_{ij} \leq y_j \quad \forall i \\ x_{ij} \in \{0, 1\}, \quad y_j \in \{0, 1\}$$

With  $d_i = 1$ ,  $C_j = 10$ ,  $x_{ij} = 1$ ,  $y_j = 0.1$  and other assignments zero satisfy the aggregate relaxation but violate the added link.

**Benchmark guidance:** Compare aggregate and linked models with identical data and objectives; verify integer feasibility, then record LP bounds, fractional openings, root gaps, and runtime.

**Constructed tasks:** 23 EfficientOpt; 3 extension.

Source: OptTips / T49.

## FAMILY D

## T50 Tensor-variable and low-rank modeling

**Core idea:** Preserve multiway data and decisions; tensor decompositions and unfolding penalties have distinct rank assumptions and approximation effects that require explicit qualification.

**Applicable patterns:** Consider tensor completion, recommendation, imaging, video, or spatiotemporal data; identify modes, observations, and plausible rank structure before selecting a representation.

**Inefficiency symptoms:** Flattening may hide mode relationships and produce dense operators, while restrictive ranks or inappropriate decomposition assumptions can exclude relevant tensor solutions.

**Expert actions:** Declare dimensions and ranks; select CP, Tucker, TT/TR, or an unfolding-based surrogate. Distinguish exact rank-restricted parameterization from changed objectives and approximate reconstruction.

**Mathematical example:** For  $\mathcal{X}, \mathcal{M} \in \mathbb{R}^{I \times J \times K}$ ,  $P_\Omega$  retains observations, zeroing others. Factors lie in  $\mathbb{R}^I, \mathbb{R}^J, \mathbb{R}^K$ , respectively; integer  $R \geq 1$ .

## Rank restricted tensor model

$$\inf_{\mathcal{X}} \frac{1}{2} \|P_\Omega(\mathcal{X} - \mathcal{M})\|_F^2 \\ \text{rank}_{\text{CP}}(\mathcal{X}) \leq R$$

## CP parameterization

$$\mathcal{X} = \sum_{r=1}^R a_r \circ b_r \circ c_r \\ \inf_{\{a_r, b_r, c_r\}} \frac{1}{2} \|P_\Omega(\mathcal{X} - \mathcal{M})\|_F^2$$

The parameterization preserves the rank restricted set but is nonconvex and may lack a minimizer. Unfolding nuclear norm penalties define different surrogates.

**Benchmark guidance:** Compare explicit tensors and CP factors with the same rank cap and mask; record reconstruction error, factor norms, memory, and initialization dependence.

**Constructed tasks:** 3 in the extension.

Source: OptTips / T50.

## Selected benchmark cases

1–4 of 8

Displayed formulas show the relevant modeling change. Recorded solver Runtime excludes model construction; objective agreement alone does not prove formulation equivalence.

## FAMILY A

## T03 Implied variable bounds

**Case:** T03\_001

**Core idea:** Expose bounds implied by precedence and conflict structure, strengthening variable domains before solving the integer model.

**Applicable patterns:** Integer energy quantities obey separation constraints; conflicting tasks require distinct operating windows with ordered binary activation indicators.

**Inefficiency symptoms:** Nonnegative quantity bounds and unrestricted binary window indicators leave useful implications implicit during search.

**Expert actions:** Propagate quantity lower bounds and fix the first activation indicators using a proven clique lower bound.

**Mathematical example:** Here  $a, b \geq 0$  are separations and  $\omega = 8$  is a verified conflict-clique size. Keep quantity upper bounds, capacities, task assignments, conflicts, activation order, and all costs in both models.

## Ordinary reference

$$\begin{aligned} x - y &\geq a, & y - z &\geq b, \\ x, y, z &\in \mathbb{Z}_{\geq 0}, \\ u_w &\geq u_{w+1} \end{aligned}$$

## Expert reference

$$\begin{aligned} x &\geq a + b, & y &\geq b, \\ u_w &= 1 & (1 \leq w \leq \omega) \end{aligned}$$

**Benchmark guidance:** Use identical energy plans and conflicts; check bound validity and objective agreement, then compare preprocessing, solver work, and Runtime.

**Recorded Runtime:** Ordinary reference: 2,619.12 s; expert: 21.20 s.  
Both OPTIMAL; matching objective values.

Technique card: T03

## FAMILY B

## T10 Pairwise assignment marginals

**Case:** T10\_011

**Core idea:** Couple pairwise product variables through assignment marginals, giving an extended representation of each pair of joint choices.

**Applicable patterns:** Departments choose one city each, with binary assignments, city capacities, and communication costs coupling selected department pairs.

**Inefficiency symptoms:** Separate product inequalities allow weak fractional combinations; binary joint variables add redundant integrality declarations.

**Expert actions:** Use continuous nonnegative joint variables and row/column marginal equations, preserving one-hot assignments, capacities, and all cost terms.

**Mathematical example:**  $p, q$  index departments and  $i, j$  cities;  $y$  represents joint assignments. With nonnegativity and one-hot marginals, each pair has its joint-choice convex hull; shared capacities remain.

## GPT-5.5

$$\begin{aligned} y_{ij}^{pq} &\leq x_{pi}, y_{ij}^{pq} \leq x_{qj}, \\ y_{ij}^{pq} &\geq x_{pi} + x_{qj} - 1, \\ x_{pi}, y_{ij}^{pq} &\in \{0, 1\} \end{aligned}$$

## Expert reference

$$\begin{aligned} \sum_j y_{ij}^{pq} &= x_{pi}, \\ \sum_i y_{ij}^{pq} &= x_{qj}, \\ x_{pi} \in \{0, 1\}, 0 &\leq y_{ij}^{pq} \leq 1 \end{aligned}$$

**Benchmark guidance:** Compare both formulations on identical assignment and interaction data; verify feasible decisions and equal objectives, then measure relaxation bounds, dimensions, and solver work.

**Recorded Runtime:** GPT-5.5: 3,693.24 s; expert: 20.75 s.  
Both OPTIMAL; matching objective values.

Technique card: T10

## FAMILY C

## T12 LP duality for absolute residuals

**Case:** T12\_004

**Core idea:** Use LP duality to exchange observation-wise residual inequalities for bounded dual variables and coefficient-wise orthogonality equations.

**Applicable patterns:** Fit three unrestricted coefficients by minimizing total absolute residual; the primal LP must be feasible with finite optimum.

**Inefficiency symptoms:** Two residual inequalities per observation create a large explicit constraint matrix.

**Expert actions:** Derive the dual with correct signs and bounds; recover the three required fitted coefficients and verify the original residuals.

**Mathematical example:**  $A$  contains three features per observation,  $y$  the responses, and  $d$  absolute-residual bounds. The primal minimizes  $\mathbf{1}^T d$ ; its dual maximizes the same value with  $-1 \leq w_i \leq 1$ .

## Ordinary reference

$$\begin{aligned} \min_{\beta \in \mathbb{R}^3, d \geq 0} & \mathbf{1}^T d \\ -d &\leq A\beta - y \leq d \end{aligned}$$

## Expert reference

$$\begin{aligned} \max_w & y^T w \\ A^T w &= 0, \quad -1 \leq w_i \leq 1 \end{aligned}$$

**Benchmark guidance:** Use identical observations for both models; verify strong-duality conditions, matching objectives, and requested coefficient recovery, then compare dimensions, construction cost, and solver work.

**Recorded Runtime:** Ordinary reference: 279.26 s; expert: 2.78 s.  
Both OPTIMAL; matching objective values.

Technique card: T12

The timing records do not establish recovery of the required primal decisions.

## FAMILY D

## T18 Network flow for tour connectivity

**Case:** T18\_001

**Core idea:** Carry one commodity from a depot to every city, enforcing connectivity through flow conservation and selected-arc capacities.

**Applicable patterns:** The task requires a minimum-cost Hamiltonian tour, with binary arcs and one incoming and outgoing selected arc per city.

**Inefficiency symptoms:** Order-variable connectivity constraints can leave weak fractional tours in the LP relaxation.

**Expert actions:** Replace order constraints with depot supply, unit city demands, and nonnegative flows bounded by selected-arc capacities.

**Mathematical example:** Both models minimize  $\sum_{i \neq j} c_{ij} x_{ij}$  and retain degree equations.  $\Delta f$  is outflow minus inflow: depot 0 supplies  $n - 1$  units, each other city consumes one, and  $f_{i0} = 0$ .

## Ordinary reference

$$\begin{aligned} 1 &\leq u_i \leq n - 1, \\ u_i - u_j + n x_{ij} &\leq n - 1 \\ (i \neq j; i, j &\neq 0) \end{aligned}$$

## Expert reference

$$\begin{aligned} 0 &\leq f_{ij} \leq (n - 1) x_{ij}, \\ \Delta f_0 &= n - 1, \quad \Delta f_i = -1 \\ (i \neq 0; f_{i0} &= 0) \end{aligned}$$

**Benchmark guidance:** Use identical cities and distances; verify one complete tour and equal objectives, then compare relaxation bounds, model dimensions, and branch-and-bound work.

**Recorded Runtime:** Ordinary reference: 249.47 s; expert: 12.97 s.  
Both OPTIMAL; matching objective values.

Technique card: T18

## Selected benchmark cases

5–8 of 8

Displayed formulas show the relevant modeling change. Recorded solver Runtime excludes model construction; objective agreement alone does not prove formulation equivalence.

## FAMILY E

## T06 Native semicontinuous domains

**Case:** T06\_010

**Core idea:** Encode zero-or-positive production directly with a native semicontinuous domain, replacing explicit activation binaries and their linking inequalities.

**Applicable patterns:** Laboratory reagent batches have positive minimum and finite maximum sizes; activation appears only in quantity links, without separate costs.

**Inefficiency symptoms:** Each product-shift pair adds one binary and two linking rows to the model.

**Expert actions:** Declare production semicontinuous with the same bounds; preserve inventory balances, shared capacities, and production and holding costs.

**Mathematical example:** Here  $q_{pt}$  is production and  $y_{pt}$  activation, with  $0 < L_{pt} \leq U_{pt}$ . Both minimize production plus inventory cost. The zero-or-interval domain retains the discrete decision.

## Ordinary reference

$$L_{pt}y_{pt} \leq q_{pt} \leq U_{pt}y_{pt}, \\ y_{pt} \in \{0, 1\}$$

## Expert reference

$$q_{pt} \in \{0\} \cup [L_{pt}, U_{pt}] \\ \text{(native semicontinuous)}$$

**Benchmark guidance:** Build both representations on identical data; verify feasible quantity plans and objective agreement, then compare variable counts, solver work, and Runtime.

**Recorded Runtime:** Ordinary reference: 411.77 s; expert: 31.68 s. Both OPTIMAL; matching objective values.

Technique card: T06

## FAMILY F

## T09 Convex piecewise-linear costs

**Case:** T09\_008

**Core idea:** Represent convex piecewise-linear costs by supporting lines, removing discrete segment choices while preserving the minimum cost.

**Applicable patterns:** Allocations have four ordered segments, nonnegative nondecreasing slopes, group quantity requirements, and shared resource limits.

**Inefficiency symptoms:** Binary filling indicators introduce unnecessary integrality and enlarge the model for a convex cost function.

**Expert actions:** Replace incremental quantities and filling indicators with one bounded quantity and one epigraph cost per unit.

**Mathematical example:** For each unit,  $b_0 = F(b_0) = 0$ ,  $\Delta_k = b_{k+1} - b_k$ ,  $\beta_k = F(b_k) - m_k b_k$ , and  $0 \leq y_k \leq \Delta_k$ . Ordinary cost is  $\sum_k m_k y_k$ ; binary  $z_k$  links segments. Minimize summed costs;  $k = 0, \dots, 3$ , with links for  $k < 3$ .

## Ordinary reference

$$x = \sum_k y_k, \\ y_k \geq \Delta_k z_k, \\ y_{k+1} \leq \Delta_{k+1} z_k$$

## Expert reference

$$c \geq m_k x + \beta_k \quad (\forall k), \\ 0 \leq x \leq b_4$$

**Benchmark guidance:** Preserve group requirements and resource capacities; verify convexity and objective agreement, then compare binary counts, matrix size, and solver work.

**Recorded Runtime:** Ordinary reference: 378.62 s; expert: 42.75 s. Both OPTIMAL; matching objective values.

Technique card: T09

## FAMILY G

## T13 Lagrangian resource dualization

**Case:** T13\_006

**Core idea:** Price the shared labor resource to separate technology blocks, then express their adjusted values through an exact LP dual.

**Applicable patterns:** Technology intensities form continuous simplices coupled by one labor budget; the primal LP must be feasible with finite optimum.

**Inefficiency symptoms:** A monolithic model retains many intensity variables despite limited coupling between blocks.

**Expert actions:** Dualize the labor limit with one nonnegative price and represent each block value using a free variable.

**Mathematical example:**  $v_{ik}, r_{ik}$  denote value and labor, and  $R$  the budget. At price  $\lambda$ , each block maximizes  $v_{ik} - \lambda r_{ik}$ ; minimizing  $R\lambda + \sum_i u_i$  gives the primal optimum.

## GPT-5.5

$$\max_{x \geq 0} \sum_{i,k} v_{ik} x_{ik} \\ \sum_k x_{ik} = 1 \quad (\forall i), \\ \sum_{i,k} r_{ik} x_{ik} \leq R$$

## Expert reference

$$\min_{\lambda, u} R\lambda + \sum_i u_i \\ u_i + r_{ik}\lambda \geq v_{ik} \quad (\forall i, k), \\ \lambda \geq 0, \quad u_i \in \mathbb{R}$$

**Benchmark guidance:** Use identical values, labor requirements, and budget; verify strong duality and objective agreement, recover requested mixing decisions, then compare model size and solver work.

**Recorded Runtime:** GPT-5.5: 1,048.04 s; expert: 4.44 s. Both OPTIMAL; matching objective values.

Technique card: T13

The timing records do not establish recovery of the required primal decisions.

## FAMILY H

## T22 Compact robust counterpart

**Case:** T22\_008

**Core idea:** Dualize each budgeted deviation problem to replace enumerated scenarios with a compact formulation of the robust feasible region.

**Applicable patterns:** Production and deviations are nonnegative; each resource row has bounded uncertainty, with integer  $\Gamma$  for the displayed subset enumeration.

**Inefficiency symptoms:** Enumerating allowed deviation subsets creates combinatorially many constraints and increases construction workload.

**Expert actions:** Dualize the uncertainty budget and component bounds with nonnegative prices; add the resulting counterpart to each resource row.

**Mathematical example:** For each row,  $a, d, b$  give nominal coefficients, deviations, and capacity;  $0 \leq z \leq 1$  and  $\sum_{j \in S} z_j \leq \Gamma$  define uncertainty. Preserve profit and other restrictions.

## Ordinary reference

$$a^T x + \sum_{j \in Q} d_j x_j \leq b \\ \forall Q \subseteq S, \quad |Q| \leq \Gamma$$

## Expert reference

$$a^T x + \Gamma \mu + \sum_{j \in S} \lambda_j \leq b, \\ \mu + \lambda_j \geq d_j x_j \quad (\forall j \in S), \\ \mu, \lambda_j \geq 0$$

**Benchmark guidance:** Use identical uncertainty sets, objectives, and production restrictions; check inner LP duality and robust-feasibility equivalence, then compare rows, construction cost, and Runtime.

**Recorded Runtime:** Ordinary reference: 157.49 s; expert: 13.63 s. Both OPTIMAL; matching objective values.

Technique card: T22

Historical execution input identity is not hash-verified for this pair.